

✅ Chapter 1: Introduction to C Programming

Conceptual Questions:

1. What is C programming?
2. Who developed the C language and when?
3. What are the features of the C language?
4. What is the structure of a C program?
5. What are the basic data types in C?
6. What is a compiler? Name a few C compilers.
7. What is the difference between a compiler and an interpreter?
8. What is a keyword? List 10 C keywords.
9. What is the purpose of the main() function?
10. What is the difference between #include<stdio.h> and <stdio.h>?

Practical/Code Questions:

1. Write a C program to print "Hello, World!".
2. Write a C program to print your name and age.
3. Identify the error in the following code:

```
main() {  
    printf("Hello");  
}
```

4. Write a program to print the size of int, float, char, and double using sizeof().

✅ Chapter 2: Variables, Constants, and Data Types

Conceptual Questions:

1. What is a variable? How do you declare one in C?
2. What are the rules for naming a variable?
3. What is the difference between int a; and const int a = 10;?
4. What is the use of #define in C?
5. What is the scope of a variable?

Code Questions:

1. Write a C program to swap two numbers using a temporary variable.
2. Write a program to declare and print all types of variables.
3. Predict the output:

```
int a = 10, b = 5;
```

```
printf("%d", a + b * 2);
```

✓ Chapter 3: Operators and Expressions

Conceptual Questions:

1. What are the different types of operators in C?
2. Explain the precedence and associativity of operators.
3. What is the difference between ++i and i++?
4. What is a ternary operator? Give an example.

Code Questions:

1. Write a program to find the average of three numbers.
2. Write a C program to check whether a number is even or odd using the modulus operator.
3. Write a program to swap two numbers without using a third variable.

✓ Chapter 4: Input and Output

Conceptual Questions:

1. What is the purpose of scanf() and printf()?
2. How do you read a character, string, and integer from the user?
3. What is the difference between getchar() and gets()?

Code Questions:

1. Write a program to take name and age from the user and display it.
2. Write a program to read two float values and print their sum.
3. Write a program to input a character and display its ASCII value.

✓ Chapter 5: Control Statements (if, if-else, switch, loops)

Conceptual Questions:

1. What is the difference between if and switch?
2. How does a while loop differ from a do-while loop?
3. What are break and continue statements?

Code Questions:

1. Write a program to find the largest of three numbers using if-else.
2. Write a program to check whether a number is positive, negative, or zero.
3. Write a menu-driven program using switch to perform arithmetic operations.
4. Write a program to print numbers from 1 to 10 using a while loop.
5. Write a program to calculate the factorial of a number using a for loop.

Chapter 6: Arrays

Conceptual Questions:

1. What is an array? How is it different from a normal variable?
2. How do you declare and initialize a one-dimensional array?
3. What is the default value of an uninitialized array?
4. How can you find the size of an array?

Code Questions:

1. Write a program to store 5 numbers in an array and print them.
2. Write a program to find the largest element in an array.
3. Write a program to find the sum and average of all elements in an array.
4. Write a program to reverse an array.
5. Write a program to count even and odd numbers in an array.
6. Write a program to copy one array into another.
7. Write a program to sort an array in ascending order.
8. Write a program to search for an element in an array (linear search).